



# Generative AI chatbots in higher education: Student experiences and perceived ethical challenges

Neda Hadinejad<sup>a,1</sup>, Katarina Sperling<sup>b</sup>, Cormac McGrath<sup>c,\*</sup>

<sup>a</sup> Non-affiliated scholar

<sup>b</sup> Department of Behavioural Sciences and Learning, Linköping University, Norrköping, Sweden

<sup>c</sup> Stockholm University, Department of Education, Frescatiweg 54, SE-106 91

## ARTICLE INFO

### Keywords:

AI chatbots  
Higher education  
Academic integrity  
Responsible AI use

## ABSTRACT

The integration of generative artificial intelligence (GAI) chatbots in higher education has introduced both opportunities and challenges for students' academic practices. To date, most qualitative research has examined the implications of this phenomenon for teaching practices and the experiences of teachers. This explorative study investigates how students use GAI chatbots in their studies and how they evaluate AI generated text, particularly in relation to academic integrity, reliability, and ethical norms. Using a qualitative research approach, we conducted semi-structured interviews with higher education students to a) examine students' experiences of using GAI chatbots in their academic studies, and b) to explore how students perceive the ethical impact of using chatbots on their studies. The findings reveal that while students use GAI chatbots to support writing, idea generation, and language improvement, they also face challenges related to plagiarism concerns, the reliability of AI-generated content, and the lack of clear institutional guidelines on responsible AI use. Additionally, the study highlights how students develop individual strategies to navigate these challenges, including seeking validation from educators and critically assessing AI-generated outputs. The findings emphasize the need for clearer academic policies and ethical frameworks to support students in making informed and responsible decisions about the use of GAI chatbots in higher education.

## 1. Introduction

The emergence of Generative Artificial Intelligence (GAI), particularly when manifested in Chatbots like ChatGPT, has brought both promise and controversy to higher education [2,30,32]. In the context of Higher Education (HE), GAI presents potentially exciting possibilities for students by generating grammatically accurate responses to complex questions [29] but have also generated concerns about how they impact teacher-student relationships [17]. The introduction of such technologies is often accompanied by grand promises such as, providing students with personalized feedback, serving as interactive discussion partners, and introducing innovative methods for explaining complex concepts [30]. However, the presence of GAI chatbots in HE also raises significant concerns regarding academic integrity, as well as ethical and practical questions related to authenticity, and equity in education [22,27,33]. As academic institutions adapt to the integration of chatbots, existing ethical dilemmas re-emerge [25]; the potential for plagiarism [14] and

the impact on students' creative writing [29]. Qualitative research on GAI is currently developing rapidly but remains in early stages of development [7]. This study investigates how higher education students engage with GAI chatbots in their academic studies. Specifically, it examines the ways students integrate these tools into their writing and learning processes, as well as the methods they use to critically evaluate accuracy and reliability of AI-generated texts. Through qualitative interviews, it examines their perceptions of ethical use of chatbots and their role in facilitating learning processes. In the context of higher education studies, it is crucial to explore how students perceive and navigate the use of chatbots for academic purposes, addressing the ethical challenges and potential impacts on their academic practices [18], not least given students increasing use of GAI in their studies [35] and the associated concerns about ethical and responsible use of AI [11] and in particular given that studies with a focus on students' perspectives situated practices are lacking [34]. This study addresses this research gap by focusing on a viewpoint that is less emphasized, and

\* Corresponding author.

E-mail address: [cormac.mcgrath@edu.su.se](mailto:cormac.mcgrath@edu.su.se) (C. McGrath).

<sup>1</sup> Independent Researcher.

underexplored in the current literature [26]. To that end, this study targets two research questions:

- What are students' experiences of using AI chatbots such as ChatGPT in their academic studies?
- How do students perceive the ethical impact of using chatbots on their studies?

This paper is structured as follows; first it identifies research on GAI chatbots' impact on students' writing, highlighting its potential to enhance academic writing through improvements in idea generation, proofreading, and editing; second, the challenges GAI chatbots pose to creativity and academic integrity, including concerns about idea ownership and plagiarism are presented. The paper then present the findings of the empirical work, before discussing the findings in light of existing research.

### 1.1. Students and GAI

GAI-chatbots, such as ChatGPT, Claude and Co-pilot are being used by students to assist in creating essays, articles, research papers, and assignments [14,35]. Such engagement raises significant concerns about academic integrity and potential AI-facilitated cheating [5]. According to Hamerman et al., [20], students utilize chatbots as versatile writing assistants to support various stages of their written assignments in diverse ways. For instance, they employ these tools to generate ideas as part of their pre-writing strategies, which helps them initiate the writing process. Students recognize AI chatbots as valuable tools for structuring assignments, generating ideas, and improving writing fluency [12]. Chatbots may also aid in producing initial drafts, enabling students to focus on critical thinking and analyzing writing topics, rather than solely on generating initial ideas. Additionally, it has been reported that students seek feedback from chatbots to enhance their writing skills. Similarly, Imran & Almusharraf, [21] highlight that ChatGPT functions as an all-encompassing tool, offering assistance from idea generation to proofreading and editing. Furthermore, its ability in generating accurate and relevant responses across various academic disciplines highlights its practicality for tasks such as drafting essays, refining arguments, and initiating research [1,29].

### 1.2. Concerns about students' use of GAI

A number of concerns have been raised in connection to students' critical thinking in conjunction with using GAI chatbots. Rudolph et al., [31], for example, warn that chatbots should not substitute critical thinking and originality but should be utilized as a tool to enhance writing and research skills. Others have expressed concerns regarding the quality of chatbot responses and the potential biases embedded in AI-generated content [9,10,32]. More recently, concerns have been raised that the claimed positive effects of using GAI chatbots, namely that they off-load cognitive load, is, in fact, one of the quintessential elements of learning and that off-loading cognitive load, may in fact jeopardize learning processes themselves [24,38]. A further concern, highlighted by Dalalah & Dalalah, [16], is that plagiarism could become more widespread as universities lack the tools to determine if a text is authored by GAI or by a student, or a hybrid of both. Bowles & Kruger, [8] believe that educators should explore alternative assessment methods beyond take-home essay which underscores the need for heightened awareness and ethical practices when integrating chatbots into academic studies. Moreover, students are struggling to understand what forms of interaction are allowed with tools which are become increasingly ubiquitous [13]. Consequently, understanding students' perceptions about ethical use of GAI is important. Technologies which mediate how students engage with learning may involve potentially unsettling consequences for academic integrity [23] by blurring the boundaries of authorship and originality [12] and raising questions

about accountability in academic work [17]. The fundamental principle of integrity involves acting responsibly in the creation of knowledge and properly acknowledging others' contributions. It also emphasizes refraining from claiming ownership of ideas, thoughts, or work that are not originally one's own [3].

### 1.3. New technologies but age-old problems

AI potentially exacerbates some of these aforementioned challenges in education, where plagiarism detection tools use AI to ensure originality, while tools like ChatGPT can generate content that is undetectable in conventional plagiarism detection tools [31]. As a result, conventional detection software designed to identify similarities between student work and existing texts proves largely ineffective, as the generated content is not directly copied from other sources. Each time a prompt is entered, the technology generates a unique response, resulting in variations that prevent similarities with other essays created using the same tool. This further renders traditional plagiarism detection software ineffective. Consequently, university teachers are unable to detect AI use, which means an understanding of students' perceptions of ethical use of GAI is imperative to enable teachers to navigate higher education teaching and learning practices, which are becoming an increasingly messy [11]. We also identify that the vast majority of policy research policy research on GAI in higher education identifies that the main responsibilities for regulating GAI use in class is often delegated to educators [15,28,37]. Consequently, students may receive conflicting information about what is considered good practice [13].

Finally, the emergence of AI chatbots has introduced new challenges to maintaining academic integrity, particularly in university assessment practices. Its presence has increased teachers' criticality and suspicion, diminishing their trust in students' work. This skepticism, often rooted in the potential misuse of chatbots, influences perceptions of text quality, leading to more stringent assessments and a tendency to downgrade, as teachers adapt their assessments in response to this evolving academic landscape [17]. Blum, [6] points out that there is not a universal understanding of academic integrity among different academic groups and student bodies. Kroes and Verbeek [23] account for "*Technical artefacts actively influencing their users, changing the way they perceive the world and the way they interact with each other*" (2014, p. 1). Recognizing these perspectives can help educators and policymakers develop AI guidelines that balance its benefits with responsible use, ultimately supporting effective learning outcomes [12]. Accordingly, having a better understanding of students' ways of engaging with these tools is necessary.

## 2. Method

This study employs semi-structured interviews to address our research questions. The qualitative nature of the study enabled an exploration of how students navigate the complexities of ethical decision-making in their interactions with GAI chatbots. Moreover, the conversational format of the interviews encouraged participants to reflect deeply on their motivations and uncertainties, allowing for rich data collection. Participants were purposefully selected to ensure a variety of age, experience and potential perspectives, (see Table 1 below). Invitations were sent to master's students enrolled in an International Master of Educational Research program at the University of Gothenburg, as well as students from the Computer Science, in total, 12 students participated: 10 from the IMER program and 2 from Computer Science. Interviews were conducted online, lasting between 30 and 60 min, and addressed issues such as academic integrity, ethical dilemmas, and the role of chatbots in learning processes (see Appendix 1 for the list of interview questions).

Informed consent was obtained, with participants receiving detailed information about the study's objectives, their rights, and the measures in place to protect their anonymity. Pseudonyms were assigned, and

**Table 1**  
Participants.

| Participants   |                       |    |
|----------------|-----------------------|----|
| Gender         | Female                | 11 |
|                | Male                  | 1  |
| Age Range      | 25–30                 | 6  |
|                | 30–35                 | 1  |
|                | 35–40                 | 3  |
|                | 40–45                 | 2  |
| Academic Level | Master                | 12 |
| Field of Study | Educational Research  | 11 |
|                | IT and Learning       | 1  |
| University     | Gothenburg University | 12 |

data were securely stored with restricted access, complying with ethical standards in educational research [3]. Additional steps were taken to ensure transparency throughout the research process, including opportunities for participants to review and clarify their responses if desired.

### 2.1. Data analysis

Following the qualitative content analysis procedure outlined by Graneheim and Lundman [19], we processed the data in the following ways; i) the data was transcribed and read repeatedly to get a sense of meaning, ii) meaning units were identified by identifying parts of the text (words, sentences, or paragraphs) iii) condensed meaning units were identified, where the text was shortened while preserving the core meaning, iv) codes were assigned to the condensed meaning units, v) Codes were then sorted into categories, vi) themes were formed across categories, vii) ensured trustworthiness across the themes were ensured through a process of negotiated consensus among the research team members. NH did step i, NH and CMG did steps ii-vii and KS engaged in steps vi and vii specifically. Each step of the process entailed interpretative and iterative work aimed at abstracting meaning in relation to the research questions. Ambiguities in coding or categorisation were discussed in-depth, often across multiple stages of the analysis, to strengthen the robustness of the findings. Assigning meaning units to exclusive categories required constant movement between transcripts and codes to reflect participants' sentiments accurately. We present an example in Table 2 below where we have a brief overview of the coding process.

(see Appendix 2 for an example of coding overview). The naming of themes was revised several times to ensure alignment between the inductive analysis, the underlying categories, and the guiding research questions. Throughout the process, we remained mindful of our dual positions as researchers and educators, seeking to avoid imposing pre-conceptions about AI or its potential value in education. To further minimize bias, all interview texts were anonymized, thereby preventing prior knowledge of participants from shaping the interpretation of their perspectives.

## 3. Findings

Here, we present the study's four key themes: *Learning support*, *Creativity*, *authorship and critical thinking*, *Plagiarism and academic honesty* and *Evolving ethical norms*. The first two themes align with the first research question, and the other two themes align with the second

research question.

### 3.1. Learning support

Respondents identified that AI chatbots was useful to them in their learning, through different forms of interaction and engagement. Many reported that ChatGPT significantly enhanced their learning process by correcting their texts and improving their academic tone:

*"I use ChatGPT to polish my paragraphs. Every time I finished writing something I will copy and pasted to ChatGPT and ask it to polish this paragraph or text and it is amazing."*

This was particularly useful when learning a new language, as explained by one of the respondents:

*"I use Chat GPT to translate any sort of Swedish words or phrases... when we were learning English at early age, there were those dictionaries, like the Oxford elementary dictionary. The definitions that we used to learn from those books were quite easy to understand, so as ChatGPT. It's easy to do the same when I wanted to learn Swedish, and I asked it to give me easy definitions in Swedish. So, it was like a personalized learning coach for me studying Swedish language."*

As reflected in the interviews, respondents who were non-native speakers, in particular, appreciated the language support provided by the tool, citing improvements in grammar, vocabulary, and the overall formality and academic tone of their writing in the assignments as a fundamental part of their learning. However, respondents also highlighted the chatbots tendency to use a limited or repetitive set of words, which they felt was not particularly helpful for expanding academic English vocabulary:

*"I use Word Tune to see if my academic writings have any grammatical errors, mistakes and just to double check stuff... since ChatGPT is giving a correction from its experience, piled experience, it tends to give repeated vocabulary that the readers know that it is actually not from a human. So, it tends to make all of the writings look similar. In my opinion, it is better for me to write on my own and then so that's my concern."*

One respondent reported using ChatGPT's flaw, particularly its tendency to fabricate citations, as a way to discover relevant academic sources:

*"Here's an example of how I use ChatGPT... So, I ask, here's my key words: teachers, digital competence, South Korea, la, la, la, ... Can you give me a good example of reference? Then, if that catches up to give me that title of reference, then I copy and paste it and try to go to the links. If GPT actually gave me the reference, which actually exists, ... If GPT actually gave me the real existing thing, then I use that and I read that abstract by myself and I don't even ask ChatGPT to summarize it or give me the idea of what the article is about."*

This example illustrates how students creatively adapt to the limitations of AI tools by developing their own verification strategies. Instead of relying on AI-generated references, this respondent used ChatGPT to generate possible citations and then cross-checked them against real academic databases to find valid sources.

**Table 2**  
An example of the coding process and theme development.

| Meaning Unit                                                             | Condensed Meaning Unit                            | Code                | Subcategory      | Category           | Subtheme                         | Theme                                                 |
|--------------------------------------------------------------------------|---------------------------------------------------|---------------------|------------------|--------------------|----------------------------------|-------------------------------------------------------|
| "I use ChatGPT to polish my paragraphs... it is amazing."                | ChatGPT improves writing clarity and tone         | Language refinement | Language support | Experience with AI | Writing assistance               | Theme 1: Learning Support                             |
| "ChatGPT offers useful starting points, but I must refine its output..." | ChatGPT stimulates but cannot replace originality | Idea stimulation    | Creativity       | Experience with AI | Balancing automation with effort | Theme 2: Creativity, Authorship and Critical Thinking |

### 3.2. Creativity, authorship and critical thinking

In relation to the use of AI chatbots the respondents often reflected upon the impact of chatbots on their creativity and authorship. While some valued GAI chatbots for stimulating new ideas and perspectives, others worried that excessive reliance might suppress originality, as expressed by one respondent:

*“ChatGPT offers useful starting points, but I must refine its output to ensure the work remains truly mine... I take my text and paste it into ChatGPT, and it rephrases it instead. It actually gives me the words that I don't usually use... occasionally yes, I keep them, but most of the words are words that I never use. So, I don't (keep them). I am not sure of the meaning of the words and then I guess, No! I cannot stand back. I cannot take pride in using those words because they are not mine. So, I alter the words.”*

The respondents reported actively evaluating each situation to determine what constitutes genuine authorship and how to maintain academic integrity when receiving linguistic suggestions. In such cases, they may struggle to take pride in phrasing that does not reflect their own voice or choice of words.

Respondents also reflected upon the limitations that affect their trust in GAI chatbots. This relates to the fact of lack of transparency in generating response and GAI's risk of “hallucinating” answers. One respondent stated:

*“ChatGPT offers answers without explaining the reasoning, so I always double-check its output.”*

This statement shows a constant workaround from their end, validating the answers that are generated. However, more than reflecting on their personal use, respondents often return to the societal dangers of AI chatbots and the risks of a decline in independent thinking and intellectual engagement. Some reported that excessive reliance on chatbots can hinder creativity and, more broadly, is not beneficial for society. For example:

*“I don't think that it is good for society. Since people's knowledge is getting lower, and we are just relying on AI to have all the knowledge which means that individuals become more stupid, which I don't like.”*

This perspective reflects a broader concern that dependency on GAI chatbots may lead to a decline in independent thinking and intellectual engagement. By outsourcing critical cognitive tasks to AI students seem aware that GAI chatbots can turn us into passive consumers of information rather than active creators of knowledge. In this regard, also students raise concerns related to the long-term impact of chatbots on creativity, problem-solving abilities, and the overall development of intellectual activities in society.

### 3.3. Plagiarism and academic honesty

Closely aligned to the question of authorship is the question of plagiarism, which we identify as a prominent concern. Respondents often came back to their uncertainty about how to integrate GAI-generated content without breaching academic integrity:

*“Using ChatGPT without reworking its output feels like plagiarism, yet it is sometimes hard to tell where the line is drawn.”*

While respondents recognize that directly copying AI-generated text may constitute plagiarism, they reported that the challenge lies in determining the extent to which modifications are necessary, in order to make the work their own. Ambiguity surrounding authorship and originality in the context of chatbots creates uncertainty, leaving respondents unsure about whether paraphrasing AI-generated content or using it as a starting point is ethically acceptable. Another respondent states:

*“Definitely these tools of paraphrasing are useful, but I'm telling you I'm really very cautious in using it for paraphrasing because it's tricky and it could lead you to be caught for plagiarism. Or to be caught or accused of stealing ideas.”*

These statements further illustrate the tension respondents experience when using GAI chatbots for academic studies, particularly regarding plagiarism and academic honesty. While they acknowledge the usefulness of AI-powered tools for paraphrasing, they also express significant caution, fearing that improper use could lead to accusations of plagiarism or intellectual theft. The concern is not just about copying but also about unintentionally crossing ethical boundaries due to the way AI rephrases content. Another expressed concerns about academic honesty while describing their use of chatbots in their studies:

*“I use ChatGPT for grammar check and fluency in writing. I do it since I write my assignments in English. I use it because English is not my mother tongue. But I am kind of afraid to use it. If I wrote my essays, I knew that the work is my own. I haven't stolen it from anybody but using ChatGPT and have different arrangements of the words might end up doing plagiarism, even though I'm not aware of it. So, I only use it for grammar, but not so much.”*

Respondents highlight an additional layer of complexity for non-native English speakers, who rely on ChatGPT to improve grammar and fluency. Although the tool helps enhance the clarity and readability of their writing, some students worry that using chatbots for language refinement could unintentionally alter their original ideas, making them vulnerable to plagiarism accusations. This concern reflects the broader uncertainty around what constitutes ethical AI use in academic work and underscores the need for clearer guidelines on responsible AI-assisted writing. Additionally, respondents highlighted the issue of fabricated citations, noting instances where ChatGPT generated references for non-existent sources.

### 3.4. Evolving ethical norms

As a consequence of the many concerns of GAI chatbots respondents often came back to the need to revise traditional definitions of authorship and originality. Questions emerged regarding the appropriate acknowledgment of chatbots contributions. One asked:

*“If I use ChatGPT to enhance my writing, how much of the work is still mine? Should AI be credited in any way?”*

Respondents seem to be aware the technical configurations of GAI chatbots and the fact that extensive datasets that underpin large language models are largely derived from texts written in the past. This means that the information may not reflect current perspectives or the most recent developments. For example, because, at the time of this study was conducted, ChatGPT's training data is based on older material, so its outputs cannot be reliably used as academic references, which limits its legitimacy as a source in scholarly work. These issues contribute to evolving ethical norms in the context of GAI technology, prompting academic institutions and researchers to rethink standards of accuracy, originality, and proper citation in the context of AI-generated content.

*“It is very dangerous, that we might think that what we have found on the ChatGPT is actually our own ideas! that actually is somebody's work who has done this for example. So, I think it is unethical because we don't see the resources ChatGPT was trained on.”*

Respondents expressed concerns regarding the data used to train GAI systems. They questioned whether the training data includes all personal messages, such as those on Facebook walls, or if it is limited to sources like Wikipedia. While GAI is trained on a vast amount of data, it primarily uses text, even though images may contain metadata that could be relevant. Furthermore, respondents were worried that personal data

entered into chatbots, or files uploaded for analysis in newer GPT versions might be used for future training, especially given that user feedback plays an important role in refining responses.

*“There are also some concerns about using generative AI which is for example, this data that we feed generative AI with. It’s personal, it could be personal data. So, I don’t know how companies use this data. Do they use it for commercial purpose, or do they use it for more civilians? I don’t know how they use it. So here we have like also other concerns about privacy and ethical data practices and the potential bias technology may bring.”*

These concerns highlight the uncertainty surrounding data privacy and the ethical use of personal information by AI companies. They also emphasize the urgent need for greater transparency in how user data is handled and for stronger safeguards to protect individuals’ rights in the development and deployment of generative AI systems.

Social stigma also influenced perceptions; some respondents hesitated to disclose their use of chatbots, fearing negative judgments about their academic abilities.

*“And I even sometimes tell the teachers that I want to use ChatGPT to get some ideas. And before that, I usually ask them what is your stance on this matter? Are you okay with this? I usually describe what I want to ask ChatGPT. I don’t just say I want to use ChatGPT, I told the teacher that I’m going to use it in this way. Which has proven quite successful. They usually agree with that. And the challenge the name of ChatGPT brings, that was one.”*

This account illustrates how the social context and associated stigma surrounding chatbots use compel respondents in their student role to negotiate their disclosure strategies carefully, highlighting the broader ethical and reputational concerns at play in academic environments.

#### 4. Discussion

In this study, we explored how students use AI chatbots in their academic studies and their perceptions of its ethical implications. In response to the first research question, “How do students use AI chatbots in their academic studies?”, respondents reported using ChatGPT for idea generation, structuring assignments, paraphrasing, and improving grammar and fluency, particularly for non-native English speakers. These uses suggest that students increasingly view chatbots not only as auxiliary writing tools but potentially as personalized learning assistants, helping them overcome linguistic and structural challenges in academic work. Furthermore, the challenges identified by respondents align with broader concerns in the field. Recent work on responsible AI in education highlights that without robust governance and ethical oversight, issues such as algorithmic bias and opaque data practices may undermine trust in AI tools [10]. Importantly, non-native English speakers demonstrated higher usage of ChatGPT for academic writing compared to native speakers, emphasizing its role as a valuable linguistic support tool. In contrast, native speakers, who typically have a stronger intuitive command of the language, exhibit lower usage rates, indicating a lesser need for such assistance [4]. These findings complement previous research by Hamerman et al. [20] and Imran and Almusharraf [21], who highlight the importance of critical engagement with GAI-generated content. Our study further develops this by showing that students not only recognize the need for critical evaluation but also actively adapt their strategies to manage the limitations of chatbot outputs. This adaptive use suggests that students are not passive consumers but engage in selective and critical interactions with AI chatbots, highlighting the emergence of new forms of digital literacy that combine technological use with critical thinking. We identify that the findings for RQ1 align with existing research on GAI in higher education. Still, some novel insights emerged, revealing alternative ways students engage with AI tools and social perceptions influencing its use. One finding that stands out in particular was when respondents were using GAI to

identify spurious references. The findings indicate that while students increasingly use chatbots such as ChatGPT to enhance their learning, they simultaneously encounter significant challenges concerning academic integrity, output reliability, and the evolving ethical landscape of AI use [11].

Regarding the second research question, “How do students perceive the ethical implications of using AI chatbots?”, respondents expressed concerns about plagiarism, authorship, and fabricated citations. This was illustrated by concerns about the integrity of their own ideas where respondents reflected on the autonomy of their own texts in a context where GAI is more widespread, where some respondents were worried that GAI might offset their own thinking. Some were uncertain about how much they needed to modify AI-generated content to avoid academic misconduct. Cotton et al. [14] and Dalalah & Dalalah [16] warn that AI-generated content may be difficult to detect using traditional plagiarism detection tools, and our findings support the idea that students are aware of and struggle with this ambiguity. Similarly, Corbin et al. [13] report that students struggle to determine when AI-assisted writing crosses the line into academic misconduct, mirroring participants’ concerns about maintaining integrity in AI-supported academic work. Additionally, the issue of fabricated citations, as highlighted by Lo [25], was a significant concern for participants. Students developed personal strategies to validate information, such as independently cross-checking references and these processes also suggest students wish to engage ethically with the academic studies. This illustrates an important shift; while AI tools may lower barriers to information access, they also demand a higher level of fact-checking and source validation from users. Rather than replacing critical thinking, these tools may, in fact, require more critical engagement from students, not less. This finding aligns with recent research on the impact of GAI on knowledge workers [24], where validation of AI output was seen as a skill set that needed to be adopted in future students and knowledge workers. It also highlights the need for strong critical literacy skills when using AI chatbots, as well as the importance of teaching students how to independently validate information produced by these systems [36]. This findings in this study suggest that understanding how student try and engage critically and ethically is an important area of future research [11].

Respondents in this study also reflected on broader questions of intellectual ownership and originality. As AI-generated texts become increasingly sophisticated, traditional concepts of authorship may be challenged. Participants’ concerns about whether and how to credit AI contributions suggest that academic institutions need to rethink norms around acknowledgment, originality, and intellectual contribution. Furthermore, the social stigma attached to chatbot use, particularly among students concerned about being perceived as dishonest, highlights the need for universities to openly discuss the responsible use of AI technologies in education. More broadly, these findings potentially reflect a critical shift in the skills and competencies required for academic success. Critical engagement, ethical awareness, and independent verification are becoming as important as traditional writing skills. Students are not only navigating technological tools but also a changing moral landscape in education. The traditional focus on producing original content is now intertwined with the need to responsibly manage AI-assisted work, calling for a new form of academic integrity that accounts for the role of emerging technologies, and this may also have implications for teaching and learning practices, which is quintessentially knowledge work [24].

From an institutional perspective, the findings suggest that institutions may need clearer, more transparent guidelines on AI use. This assertion is also supported by the emerging work on GAI policy which suggests both teachers and students are in need of clarity on what constitutes appropriate and ethical use of GAI [15,28,37]. Current policies may risk leaving students uncertain about what constitutes ethical behavior when using AI chatbots. Universities might consider developing AI literacy initiatives that help students use these technologies

thoughtfully, with guidance on appropriate citation practices, critical evaluation of AI-generated content, and awareness of the tools' limitations and potential biases. Moreover, this study suggests that reliance on AI chatbots has both empowering and risky aspects. On one hand, students, especially non-native English speakers, find valuable support in improving academic writing skills. On the other hand, the risks of fabricated information, diminished originality, and academic misconduct are real and growing. Higher education institutions face a pressing challenge; harnessing the benefits of AI while cultivating critical, independent thinking among students. Accordingly, it is important that educators, policymakers, and students collaboratively redefine academic integrity, creativity, and responsibility in ways that support and develop students' learning in the context of AI-supported learning. While these insights shed light on important patterns and implications, it is necessary to acknowledge the study's limitations before considering its broader applicability.

#### 4.1. Limitations of the study

The study has several limitations. First, the sample size was small (12 respondents) and drawn from a single university, limiting the generalizability of the findings. At the same time, we identify that students may identify patterns of behaviour and be able to transfer the findings of this study to their context. Similarly, teachers may be able to use the findings here to understand how students engage but also struggle with GAI in their studies. A larger, more diverse sample, including students from different institutions, disciplines, and educational levels, would provide a broader perspective on AI chatbots use and ethical concerns. Additionally, most participants were from educational research backgrounds, which may have influenced their views on AI ethics. Students in other fields, such as STEM or business studies or computer science, may engage with chatbots differently, particularly in disciplines where AI-generated content plays a more technical or analytical role. Second, self-selection bias may have influenced the study results. Participants voluntarily chose to take part in the study, meaning those with strong opinions or personal experiences with chatbots may have been more likely to participate. This could result in a skewed representation of student experiences, potentially amplifying concerns about academic integrity or ethical dilemmas while underrepresenting more neutral or indifferent perspectives. Third, the study relied on self-reported data collected through semi-structured interviews. While interviews provided qualitative insights, they are inherently subjective and dependent on participants' willingness and ability to accurately recall their experiences. Some students may have downplayed or exaggerated their use of AI chatbots due to concerns about academic honesty, while others may have lacked a full understanding of how AI functions, influencing their responses. Finally, the study took place in a specific academic and technological context, during a time when AI policies in higher education were still evolving. As institutions continue to develop clearer guidelines and as AI chatbots improve, students' perceptions and engagement with chatbots may change. Future research should revisit these questions to examine whether students' ethical concerns persist or shift over time. Despite these limitations, this study provides an understanding of student experiences with GAI chatbots, emphasizing the need for ongoing dialogue between students, educators, and policymakers to navigate the evolving ethical landscape of AI chatbots in academia.

#### CRedit authorship contribution statement

**Neda Hadinejad:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Katarina Sperling:** Writing – review & editing, Validation, Supervision, Methodology. **Cormac McGrath:** Writing – review & editing, Validation, Supervision, Methodology,

Investigation.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.caeo.2025.100311](https://doi.org/10.1016/j.caeo.2025.100311).

#### References

- [1] Ali D, Fatemi Y, Boskabadi E, Nikfar M, Ugwuoke J, Ali H. ChatGPT in teaching and learning: a systematic review. *Educ Sci* 2024;14(6):6. <https://doi.org/10.3390/educsci14060643>.
- [2] Al-Zahrani AM. The impact of generative AI tools on researchers and research: implications for academia in higher education. *Innov Educ Teach Int* 2024;61(5): 1029–43. <https://doi.org/10.1080/14703297.2023.2271445>.
- [3] Ashford T. App-centric students and academic integrity: a proposal for assembling socio-technical responsibility. *J Acad Ethics* 2021;19(1):35–48. <https://doi.org/10.1007/s10805-020-09387-w>.
- [4] Baek C, Tate T, Warschauer M. ChatGPT seems too good to be true": college students' use and perceptions of generative AI. *Comput Educ: Artif Intell* 2024;7: 100294. <https://doi.org/10.1016/j.caeai.2024.100294>.
- [5] Bin-Nashwan SA, Sadallah M, Bouteraa M. Use of ChatGPT in academia: academic integrity hangs in the balance. *Technol Soc* 2023;75:102370. <https://doi.org/10.1016/j.techsoc.2023.102370>.
- [6] Blum SD. What it means to Be a student today. In: Bretag T, editor. *Handbook of academic integrity*. Springer; 2015. p. 1–20. [https://doi.org/10.1007/978-981-287-079-7\\_57-2](https://doi.org/10.1007/978-981-287-079-7_57-2).
- [7] Bond M, Khosravi H, De Laat M, Bergdahl N, Negrea V, Oxley E, Pham P, Chong SW, Siemens G. A meta systematic review of artificial intelligence in higher education: a call for increased ethics, collaboration, and rigour. *Int J Educ Technol High Educ* 2024;21(1):1–41. <https://doi.org/10.1186/s41239-023-00436-z>.
- [8] Bowles DC, Kruger JS. Generating employable, intelligent graduates in a world with generative AI: thoughts for educators. *Pedagogy Health Promot* 2023;9(2): 75–7. <https://doi.org/10.1177/23733799231175171>.
- [9] Bringula R. ChatGPT in a programming course: benefits and limitations. *Front Educ* 2024;9. <https://doi.org/10.3389/educ.2024.1248705>.
- [10] Cerratto-Pargman T, McGrath C, Milrad M. Towards responsible AI in education: challenges and implications for research and practice. *Comput Educ: Artif Intell* 2024;100345. <https://doi.org/10.1016/j.caeai.2024.100345>.
- [11] Cerratto-Pargman T, McGrath C, Milrad M. Towards responsible AI in education: challenges and implications for research and practice. *Comput Educ: Artif Intell* 2024;100345. <https://doi.org/10.1016/j.caeai.2024.100345>.
- [12] Chan CKY, Hu W. Students' voices on generative AI: perceptions, benefits, and challenges in higher education. *Int J Educ Technol High Educ* 2023;20(1):1–18. <https://doi.org/10.1186/s41239-023-00411-8>.
- [13] Corbin T, Dawson P, Nicola-Richmond K, Partridge H. Where's the line? It's an absurd line": towards a framework for acceptable uses of AI in assessment. *Assess Eval High Educ* 2025;1–13. <https://doi.org/10.1080/02602938.2025.2456207>.
- [14] Cotton DRE, Cotton PA, Shipway JR. Chatting and cheating: ensuring academic integrity in the era of ChatGPT. *Innov Educ Teach Int* 2024;61(2):228–39. <https://doi.org/10.1080/14703297.2023.2190148>.
- [15] Dabis A, Csáki C. AI and ethics: investigating the first policy responses of higher education institutions to the challenge of generative AI. *Humanit Soc Sci Commun* 2024;11:1006. <https://doi.org/10.1057/s41599-024-03526-z>.
- [16] Dalalah D, Dalalah OMA. The false positives and false negatives of generative AI detection tools in education and academic research: the case of ChatGPT. *Int J Manag Educ* 2023;21(2):100822. <https://doi.org/10.1016/j.ijme.2023.100822>.
- [17] Farazouli A, Cerratto-Pargman T, Bolander-Laksov K, McGrath C. Hello GPT! goodbye home examination? An exploratory study of AI chatbots impact on university teachers' assessment practices. *Assess Eval High Educ* 2024;49(3): 363–75. <https://doi.org/10.1080/02602938.2023.2241676>.
- [18] Farhi F, Jeljel R, Aburezeq I, Dweikat FF, Al-shami SA, Slamene R. Analyzing the students' views, concerns, and perceived ethics about chat GPT usage. *Comput Educ: Artif Intell* 2023;5:100180. <https://doi.org/10.1016/j.caeai.2023.100180>.
- [19] Graneheim UH, Lundman B. Qualitative content analysis in nursing research: concepts, procedures and measures to achieve trustworthiness. *Nurse Educ Today* 2004;24(2):105–12. <https://doi.org/10.1016/j.nedt.2003.10.001>.
- [20] Hamerman EJ, Aggarwal A, Martins C. An investigation of generative AI in the classroom and its implications for university policy. *Qual Assur Educ* 2024. <https://doi.org/10.1108/QAE-08-2024-0149>. ahead-of-print(ahead-of-print).
- [21] Imran M, Almusharraf N. Analyzing the role of ChatGPT as a writing assistant at higher education level: a systematic review of the literature. *Contemp Educ Technol* 2023;15(4):1–14. <https://doi.org/10.30935/cedtech/13605>.
- [22] Khalil M, Er E. Will ChatGPT get you caught? Rethinking of plagiarism detection. In: Zaphiris P, Ioannou A, editors. *Learning and collaboration technologies*.

- Switzerland: Springer Nature; 2023. p. 475–87. [https://doi.org/10.1007/978-3-031-34411-4\\_32](https://doi.org/10.1007/978-3-031-34411-4_32).
- [23] Kroes P, Verbeek P-P. The moral status of technical artefacts, 17. Netherlands: Springer; 2014. <https://doi.org/10.1007/978-94-007-7914-3>.
- [24] Lee HP, Sarkar A, Tankelevitch L, Drosos I, Rintel S, Banks R, Wilson N. The impact of generative AI on critical thinking: self-reported reductions in cognitive effort and confidence effects from a survey of knowledge workers. In: Proceedings of the 2025 CHI conference on human factors in computing systems; 2025. p. 1–22.
- [25] Lo CK. What is the impact of ChatGPT on education? A rapid review of the literature. *Educ Sci* 2023;13(4):4. <https://doi.org/10.3390/educsci13040410>.
- [26] McGrath C, Farazouli A, Cerratto-Pargman T. Generative AI chatbots in higher education: a review of an emerging research area. *High Educ* 2025. <https://doi.org/10.1007/s10734-024-01288-w>.
- [27] Michel-Villarreal R, Vilalta-Perdomo E, Salinas-Navarro DE, Thierry-Aguilera R, Gerardou FS. Challenges and opportunities of generative AI for higher education as explained by ChatGPT. *Educ Sci* 2023;13(9):856.
- [28] Moorhouse BL, Yeo MA, Wan Y. Generative AI tools and assessment: guidelines of the world's top-ranking universities. *Comput Educ Open* 2023;5:100151. <https://doi.org/10.1016/j.caeo.2023.100151>.
- [29] Niloy AC, Akter S, Sultana N, Sultana J, Rahman SIU. Is Chatgpt a menace for creative writing ability? An experiment. *J Comput Assist Learn* 2024;40(2): 919–30. <https://doi.org/10.1111/jcal.12929>.
- [30] Rahman MM, Watanobe Y. ChatGPT for Education and Research: opportunities, threats, and strategies. *Appl Sci* 2023;13(9):9. <https://doi.org/10.3390/app13095783>.
- [31] Rudolph J, Tan S, Tan S. ChatGPT: bullshit spewer or the end of traditional assessments in higher education? *J Appl Learn Teach* 2023;6(1):1. <https://doi.org/10.37074/jalt.2023.6.1.9>.
- [32] Ruiz-Rojas LI, Acosta-Vargas P, De-Moreta-Llovet J, Gonzalez-Rodriguez M. Empowering education with generative artificial intelligence tools: approach with an instructional design matrix. *Sustain (Basel Switz)* 2023;15(15):11524. <https://doi.org/10.3390/su151511524>. -.
- [33] Šedlbauer J, Činčera J, Slavík M, Hartlová A. Students' reflections on their experience with ChatGPT. *J Comput Assist Learn* 2024;40(4):1526–34. <https://doi.org/10.1111/jcal.12967>.
- [34] Sperling K, Stenberg C-J, McGrath C, Åkerfeldt A, Heintz F, Stenliden L. In search of artificial intelligence (AI) literacy in teacher education: a scoping review. *Comput Educ Open* 2024;6:100169. <https://doi.org/10.1016/j.caeo.2024.100169>.
- [35] Stöhr C, Ou AW, Malmström H. Perceptions and usage of AI chatbots among students in higher education across genders, academic levels and fields of study. *Comput Educ: Artif Intell* 2024;7:100259.
- [36] Velander J, Otero N, Milrad M. What is critical (about) AI literacy? Exploring conceptualizations present in AI literacy discourse. Framing futures in postdigital education: critical concepts for data-driven practices. Cham: Springer Nature Switzerland; 2024. p. 139–60.
- [37] Wang H, Dang A, Wu Z, Mac S. Generative AI in higher education: seeing ChatGPT through universities' policies, resources, and guidelines. *Comput Educ: Artif Intell* 2024;7:100326. <https://doi.org/10.1016/j.caeai.2024.100326>.
- [38] Weidlich J, Gašević D, Drachsler H, Kirschner P. ChatGPT in education: an effect in search of a cause. *J Comput Assist Learn* 2025;41(5):e70105. <https://doi.org/10.1111/jcal.70105>.